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Projection-Induced Limits of Computation:
A Meta-Epistemic Account of Information Loss Across Reality, Experience, and Formal Symbolization

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Abstract

We develop a meta-epistemic account of fundamental limits on computation that arise *prior* to axiomatic reasoning, from the representational pipeline by which embodied agents transform reality into experience and symbols. We propose a three-layer framework: (i) an objective world layer f_0, (ii) an experiential layer f_1, and (iii) a symbolic–formal layer f_2 comprising natural language, logic, and mathematics. We model the transitions $f_0 \to f_1$ and $f_1 \to f_2$ as representation maps R_1 and R_2. Under empirically motivated capacity constraints (finite bandwidth, finite resolution, and compression), these maps are typically non-injective, inducing *loss of discriminability*: distinct world states may collapse to the same symbolic state. We prove that any computation performed purely in the symbolic space f_2, modeled as transformations over symbols, cannot recover distinctions lost under the composite map $R = R_2 \circ R_1$ without additional coupling to f_0 (e.g., measurement). We clarify why the present results are not internal axiomatic theorems inside f_2, but claims about the cognitive and representational conditions that make f_2 possible and constrain its epistemic reach, with implications for symbol grounding and representation learning.

1 Introduction

Classical discussions of "limits of computation" are often framed internally within formal systems: incompleteness [3], undecidability [10], and complexity barriers (e.g., NP-completeness) [1, 6]. These results are precise, but they presuppose a symbolic layer in which computation is defined and executed.

This paper addresses a distinct, prior question: *what constraints arise from the representational pipeline that produces formal symbolization in the first place?* Our thesis is that an unavoidable limit arises from *projection-induced information loss*: before any formal computation occurs, reality must be filtered into experience and then compressed into symbols. Consequently, computation in a symbolic layer generally cannot recover or fully determine properties of the objective layer absent additional coupling (new measurement or new representational channels). This is a meta-epistemic limit about representation, not a theorem about a particular axiom system.

Contributions.

- We formalize a three-layer pipeline (f_0, f_1, f_2) and representation maps (R_1, R_2) connecting world, experience, and symbols.
- We define *information loss* as loss of discriminability via non-injectivity and prove non-recoverability results for computations that operate solely on symbolic states.

- We separate internal formal limits (within f_2) from meta-level representational limits (conditions that generate f_2), positioning the argument relative to information theory [7, 2], rate–distortion [8], the information bottleneck [9], and symbol grounding [4], with implications for AI systems that operate predominantly in symbolic or linguistic spaces.

2 Preliminaries and Notation

We distinguish three layers:

- f_0: objective reality (world states).
- f_1: experiential reality (sensory/affective/phenomenal states).
- f_2: symbolic reality (language, mathematics, formal models).

Definition 1 (World, Experience, and Symbols)**.** *Let W denote the set (or space) of world states, E the set of experiential states, and S the set of symbolic states (utterances, formulas, models). A* symbolic computation *is modeled as a transformation $T : S \to S$ (or a composition of such transformations).*

Definition 2 (Representation Maps)**.** *We define two representation maps:*

$$R_1 : W \to E \quad \textit{(world} \to \textit{experience)}, \tag{1}$$

$$R_2 : E \to S \quad \textit{(experience} \to \textit{symbols)}. \tag{2}$$

The composite representation is

$$R := R_2 \circ R_1 : W \to S. \tag{3}$$

2.1 Information Loss as Loss of Discriminability

Many notions of "information loss" exist. Here we adopt a minimal structural definition suited to epistemic limits.

Definition 3 (Loss of Discriminability)**.** *A representation map $R : W \to S$* preserves discriminability *if it is injective. If R is non-injective, there exist $w_1 \neq w_2$ such that $R(w_1) = R(w_2)$; then w_1 and w_2 are indistinguishable at the symbolic level.*

Remark 1 (Relation to Information Theory)**.** *Non-injectivity captures a necessary condition for irreversible compression. Entropy-based notions (e.g., $H(W) - H(R(W))$) require a probability model over W and are compatible with this approach [7, 2]. We focus on discriminability to express the epistemic limitation without committing to a particular stochastic model.*

3 Representation Bottlenecks as Empirical Premises

Non-injectivity of R_1 and R_2 is not an axiom of mathematics; it is an empirically motivated structural claim about finite-capacity sensing and symbolization.

Assumption 1 (Finite-Capacity Sensing and Symbolization)**.** *For an embodied agent, (i) sensory and neural encoding has bounded bandwidth and finite resolution, and (ii) symbol systems are discrete and compressive relative to the richness of experience. These constraints imply that R_1 and R_2 are typically many-to-one in practical settings.*

World → Experience (R_1). Biological sensing extracts task-relevant features under bandwidth and dynamic range constraints; distinct physical microstates can yield phenomenally similar experiences. This aligns with standard views of perception as information processing under bottlenecks [2] and with levels-of-analysis distinctions between physical implementation and representation [5].

Experience → Symbols (R_2). Natural language and formal symbol systems discretize and compress experience into tokens and compositional structures. This compression is central to the symbol grounding problem: symbols require coupling to non-symbolic processes to acquire referential content [4].

4 Projection and Non-Recoverability

4.1 Non-injectivity Propagates Through Composition

Proposition 1 (Projection-Induced Loss)**.** *If either R_1 or R_2 is non-injective, then $R = R_2 \circ R_1$ is non-injective. Hence there exist $w_1 \neq w_2$ such that $R(w_1) = R(w_2)$, and symbolic computation in S cannot discriminate w_1 from w_2.*

Proof. If R_1 is non-injective, $\exists\, w_1 \neq w_2$ with $R_1(w_1) = R_1(w_2) = e$. Applying R_2 yields $R(w_1) = R_2(e) = R(w_2)$. If R_2 is non-injective, $\exists\, e_1 \neq e_2$ with $R_2(e_1) = R_2(e_2)$. Choose w_1, w_2 with $R_1(w_1) = e_1$ and $R_1(w_2) = e_2$ (nonempty preimages); then $R(w_1) = R(w_2)$. □

4.2 Symbolic Computation Cannot Restore Lost Distinctions

We model computation in f_2 as transformations $T : S \to S$ acting on symbolic representations.

Proposition 2 (Invariance Under Equal Representations)**.** *If $R(w_1) = R(w_2)$, then for any transformation $T : S \to S$,*

$$T(R(w_1)) = T(R(w_2)).$$

Thus, purely symbolic computation cannot separate states already collapsed by R.

Proof. Equal inputs to a function yield equal outputs. □

Proposition 3 (No Left-Inverse Without Additional Information)**.** *If $R : W \to S$ is non-injective, then there is no function $I : S \to W$ such that $I \circ R = \mathrm{id}_W$.*

Proof. If R is non-injective, $\exists\, w_1 \neq w_2$ with $R(w_1) = R(w_2) = s$. If $I \circ R = \mathrm{id}_W$, then $I(s) = I(R(w_1)) = w_1$ and also $I(s) = I(R(w_2)) = w_2$, a contradiction. □

Remark 2 (Recoverability Requires Additional Coupling)**.** *To regain distinctions lost under R, an agent must refine the representation (e.g., introduce new features, sensors, or symbols) or obtain new measurements. Manipulation in S alone cannot invert a many-to-one projection.*

4.3 Illustrative Example: Apple-as-"1"

Example 1 (Quantity Projection)**.** *Let an apple world-state be a tuple*

$$w = (q, c, t, s, m, \dots),$$

where q is quantity, c color, t taste, s size, m mass, etc. A common algebraic symbolization retains only quantity:

$$R_{qty}(w) = q.$$

Then distinct apples $w_1 \neq w_2$ may satisfy $R_{qty}(w_1) = R_{qty}(w_2) = 1$, collapsing differences in $c, t, s, m, \dots$ into the same token. No purely algebraic computation on q recovers discarded attributes without additional measurement.

5 Grounding and Reference

Definition 4 (Grounding as a Constraint on Symbols)**.** *A grounding relation connects symbols to sets of possible world states. Define a grounding map*

$$G : S \to \mathcal{P}(W),$$

where $G(s)$ is the set of world states consistent with symbol s under an interpretation. If $R : W \to S$ is a representation map, a natural choice is $G(s) = R^{-1}(s)$ (the preimage). When R is non-injective, $|G(s)| > 1$ for some s, capturing underdetermination at the symbolic level.

Remark 3 (Implications for AI: Grounding and Representation Learning)**.** *For AI systems that operate predominantly over symbols or learned latent representations, non-injectivity implies that multiple underlying situations may map to the same internal state, limiting recoverability without additional signals. This connects the present framework to classical grounding concerns [4] and to representation learning perspectives that emphasize compression–relevance trade-offs [9].*

6 Why This Is Not Axiomatic Mathematics

Axiomatic mathematics operates *within* f_2: it stipulates axioms over symbolic objects and derives theorems by formal inference. Proofs and counterexamples are internal to a chosen language and semantics.

By contrast, this paper is *meta-epistemic*: it studies the conditions under which an f_2-style symbolic layer can arise from an f_0-style world via an f_1-style experiential bottleneck. The key premises (finite bandwidth, compression, discretization) are empirical-structural claims about embodied representation, not consequences of an arbitrarily chosen axiom system. Therefore, constructing an injective map inside a formal universe does not refute the thesis; it only shows what is possible given a different representational setup than the one imposed by the $f_0 \to f_1 \to f_2$ pipeline.

7 Related Work and Positioning

Our argument is conceptually adjacent to several established lines of work:

- **Information theory and compression.** Foundational results on communication and entropy [7] and rate–distortion theory [8] formalize trade-offs between fidelity and description cost.
- **Information bottleneck.** The idea that representations retain task-relevant information under compression constraints is aligned with the bottleneck perspective [9].
- **Symbol grounding.** Symbols require coupling to non-symbolic processes to achieve referential semantics [4].
- **Internal formal limits.** Classical impossibility results in logic and computation [3, 10] concern constraints *within* f_2, whereas our contribution concerns constraints on how f_2 can represent f_0 via f_1.

8 Discussion

Internal vs. representational limits. Internal limits (incompleteness, undecidability, complexity) constrain what can be proven or decided within formal systems. The present limit constrains what can be discriminated about f_0 when computation is mediated by lossy representations induced by embodied pipelines.

Scientific modeling. Scientific models are task-oriented projections designed to preserve invariants relevant to specific questions. The framework highlights why model selection is inseparable from loss: every model is a slice with explicit representational support and inevitable blind spots.

AI systems. In AI, learned representations may compress observations into latent states. If the effective representation is non-injective relative to underlying factors of interest, then purely internal computation cannot recover distinctions without additional signals (e.g., richer observations, interaction, or supervision). This makes explicit a structural constraint on reasoning from compressed representations.

9 Conclusion

We presented a layered account $f_0 \to f_1 \to f_2$ in which both sensory experience and symbolic language act as projections. Under empirically motivated capacity constraints, these projections are typically many-to-one, yielding loss of discriminability. We proved that computation performed purely in symbolic space cannot recover distinctions lost under the representation map without additional coupling to the world. This constitutes a meta-epistemic limit on computation rooted in the genesis and structure of representation and supports a clear separation between internal formal limits and representational limits relevant to AI systems.

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